Han Peng

Address: The Queen's College, High Street, Oxford, OX1 4AW | Tel: +44(0)7818051268
Email: Han.Peng@physics.ox.ac.uk | Webpage: https://ha-ha-ha-han.github.io/

EDUCATION

2014 – **Department of Physics, University of Oxford** DPhil (PhD) Reading in Condensed Matter Physics
2010 – 2014 **University of Science and Technology of China (USTC)** B.Sc. in Applied Physics, GPA: 3.73/4.3

RESEARCH

2014 – 2018 **PhD Project: Study of Electronic Properties of Quantum Materials** **Advisor: Prof Yulin Chen**
Clarendon Laboratory, Department of Physics, University of Oxford

- **Machine learning in ARPES data analysis**
  Using convolutional neural network to extract the dispersive features in the spectrum images. The project is developed using MATLAB, Python and TensorFlow
- **Twisted bilayer graphene**
  Using Angular Resolved Photo-Emission Spectroscopy (ARPES) with Micrometre resolution to study electronic properties of twisted bilayer graphene
- **Other contributions**
  Leading the development of data analysis system for spectrum image data processing.
  There are 30 users, 4 regular contributors and 10 intern developers throughout the years. The work includes algorithm optimisation, data visualisation and graphic user interface building

2017 **Summer Research Intern: Neuron Image Registration** **Advisor: Prof Kenneth Harris**
Cortical Processing Laboratory, University College London

- Designed algorithm to register ex-vivo microscopic 2D-slices with in-vivo two-photon imaging 3D-zstacks for neurons
- Developed semi-automatic MATLAB data analysis pipeline for registering 130 2D-slices with 2 z-stacks neuroimaging data sets. The pipeline includes data I/O, cell detection, registration, result visualisation and manual modification interface. Project GitHub page: https://git.io/vAjk4

2013 **Research Intern: Study of Jamming Transition in Soft Materials** **Advisor: Prof Ning Xu**
Computational and Theoretical Soft Condensed Matter Physics Laboratory, USTC

- Computationally studied the power law phenomenon in the Jamming transition of a particle system
- Analysed the performance of energy minimization algorithm in different energy potential field

COMPUTING SKILLS AND COURSES

Programming tools: MATLAB, Python, TensorFlow
Open Course: Machine Learning by Andrew Ng, CS231n by Feifei Li
Summer School: Medical Imaging Computing Summer School 2017, University College London

TEACHING

2017 – 2018 Tutor in **Mechanics**, Jesus College, University of Oxford
2016 – 2017 Tutor in **Condensed Matter Physics**, Hertford College, University of Oxford
2015 – Demonstrator (Senior Demonstrator since 2017) in **Computational Physics**, University of Oxford

AWARDS AND FUNDING

2014 – 2018 Oxford University – China Scholarship Council Scholarship, full scholarship for PhD study
2015 Arthur H. Cooke Memorial Prize awarded by Department of Physics, University of Oxford
2011 National Endeavours Scholarship awarded by Ministry of Education of China

PUBLICATIONS

Total citation: 1310 | h-index: 6 (April 2018) | Google Scholar page: https://goo.gl/vlNvHL

Graphene and other low-dimension materials

1. H.F. Yang, C. Chen, H. Wang, Z.K. Liu, T. Zhang, **H. Peng**, N. B. M. Schröter, S. A. Ekahana, J. Jiang, L. X. Yang, V. Kandyba, A. Barinov, C. Y. Chen, J. Avila, M. C. Asensio, H. L. Peng, Z. F. Liu, and Y. L. Chen "Single Crystalline Electronic Structure and Growth Mechanism of Aligned Square Graphene Sheets" ***APL Materials, APL Materials 6, 036107 (2018), featured on the cover***
2. **H. Peng**, N. B. M. Schröter, J. B. Yin, H. Wang, T.-F. Chung, H. F. Yang, S. Ekahana, Z. K. Liu, J. Jiang, L. X. Yang, T. Zhang, C. Chen, H. Ni, H. Barinov, Y. P. Chen, Z. F. Liu, H. L. Peng, Y. L. Chen "Substrate Doping Effect and Unusually Large Angle van Hove Singularity Evolution in Twisted Bi- and Multilayer Graphene" ***Advanced Materials, 29, 27 (2017)***
3. J. B. Yin†, H. Wang†, **H. Peng† (equal contribution)**, Z. J. Tan, L. Liao, L. Lin, X. Sun, A. L. Koh, Y. L. Chen, H. L. Peng, and Z. F. Liu "Selectively enhanced photocurrent generation in twisted bilayer graphene with van Hove singularity" ***Nature Communications, 7, 10699 (2016)***
4. L. Liao, H. Wang, **H. Peng**, J. B. Yin, A. L. Koh, Y. L. Chen, Q. Xie, H. L. Peng, and Z. F. Liu "van Hove Singularity Enhanced Photochemical Reactivity of Twisted Bilayer Graphene" ***Nano Letters, 15, 5585 (2015)***

Topological materials

5. J. Jiang, N. B. M. Schröter, S.-C. Wu, N. Kumar, C. Shekhar, **H. Peng**, X. Xu, C. Chen, H. F. Yang, C.-C. Hwang, S.-K. Mo, C. Felser, B. H. Yan, Z. K. Liu, L. X. Yang, and Y. L. Chen "Observation of topological surface states and strong electron/hole imbalance in extreme magnetoresistance compound LaBi" ***Physical Review Materials, 2, 024201 (2018)***
6. A. J. Liang, J. Jiang, M. X. Wang, Y. Sun, N. Kumar, C. Shekhar, C. Chen, **H. Peng**, C. W. Wang, X. Xu, H. F. Yang, S. T. Cui, G. H. Hong, Y. Y. Y. Xia, S. K. Mo, Q. Gao, X. J. Zhou, L. X. Yang, C. Felser, B. H. Yan, Z. K. Liu, Y. L. Chen 2017. Observation of the topological surface state in the nonsymmorphic topological insulator KHgSb. ***Physical Review B, 96(16), p.165143 (2017)***
7. J. Jiang, Z. K. Liu, Y. Sun, H. F. Yang, R. Rajamathi, Y. P. Qi, L. X. Yang, C. Chen, **H. Peng**, C.-C. Hwang, S. Z. Sun, S.-K. Mo, I. Vobornik, J. Fujii, S. S. P. Parkin, C. Felser, B. H. Yan, Y. L. Chen "Signature of type-II Weyl semimetal phase in MoTe2" ***Nature Communications, 8, 13973 (2017)***
8. Z. K. Liu, L. X. Yang, Y. Sun, T. Zhang, **H. Peng**, H. F. Yang, C. Chen, Y. Zhang, Y. F. Guo, D. Prabhakaran, M. Schmidt, Z. Hussain, S.-K. Mo, C. Felser, B. Yan and Y. L. Chen "Evolution of the Fermi surface of Weyl semimetals in the transition metal pnictide family" ***Nature Materials, 15, 27 (2016)***
9. L. X. Yang, Z. K. Liu, Y. Sun, **H. Peng**, H. F. Yang, T. Zhang, B. Zhou, Y. Zhang, Y. F. Guo, M. Rahn, D Prabhakaran, Z. Hussain, S.–K. Mo, C. Felser, B. Yan and Y. L. Chen "Weyl Semimetal Phase in non-Centrosymmetric Compound TaAs" ***Nature Physics, 11, 728 (2015)***
10. Z. K. Liu, J. Jiang, B. Zhou, Z. J. Wang, Y. Zhang, H. M. Weng, D. Prabhakaran, S. -K. Mo, **H. Peng**, P. Dudin, T. Kim, M. Hoesch, Z. Fang, X. Dai, Z. X. Shen, D. L. Feng, Z. Hussain, Y. L. Chen. "A Stable Three-dimensional Topological Dirac Semimetal Cd3As2" ***Nature Materials, 13, 677 (2014)***

Other topics

11. S. L. Zhang, W. W. Wang, D. M. Burn, **H. Peng**, H. Berger, A. Bauer, C. Pfleiderer, G. van der Laan, and T. Hesjedal " Manipulation of Skyrmion Motion by Magnetic Field Gradients" ***Nature Communications, accepted***

TALKS AND PRESENTATIONS

2018.03 **Band evolution with angle in twisted bilayer graphene,** *APS March meeting 2018, Los Angeles, CA, US*
2017.11 **Register 2D neuron image with 3D,** *Neuromics project annual retreat, Washington D.C, US*

REFERENCE

Prof Yulin Chen	*Department of Physics, University of Oxford*	Yulin.Chen@physics.ox.ac.uk
Prof Thorsten Hesjedal	*Department of Physics, University of Oxford*	Thorsten.Hesjedal@physics.ox.ac.uk
Prof Kenneth Harris	*Institute of Neurology, University College London*	Kenneth.Harris@ucl.ac.uk
Prof Zhongkai Liu	*School of Physical Science and Technology, ShanghaiTech University and CAS-Shanghai Science Research Center*	Liuzhk@shanghaitech.edu.cn